

Package ‘broadcast’

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Title Broadcasted Array Operations Like 'NumPy'

Version 0.1.6

Description Implements efficient 'NumPy'-like broadcasted operations for atomic and recursive arrays.

In the context of operations involving 2 (or more) arrays,

“broadcasting” refers to efficiently recycling array dimensions without allocating additional memory.

Besides linking to 'Rcpp',

'broadcast' does not use any external libraries in any way;

'broadcast' was essentially made from scratch and can be installed out-of-the-box.

The implementations available in 'broadcast' include, but are not limited to, the following.

1) Broadcasted element-wise operations on any 2 arrays;

they support a large set of

relational, arithmetic, Boolean, string, and bit-wise operations.

2) A faster, more memory efficient, and broadcasted abind-like function,
for binding arrays along an arbitrary dimension.

3) Broadcasted ifelse-like, and apply-like functions.

4) Casting functions,

that cast subset-

groups of an array to a new dimension, cast nested lists to dimensional lists, and vice-versa.

5) A few linear algebra functions for statistics.

The functions in the 'broadcast' package strive to minimize computation time and memory usage

(which is not just better for efficient computing, but also for the environment).

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BugReports <https://github.com/tony-aw/broadcast/issues/>

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Contents

aaa00_broadcast_help	2
aaa01_broadcast_operators	4
aaa02_broadcast_casting	7
acast	10
bc.b	12
bc.bit	13
bc.cplx	15
bc.d	16
bc.i	17
bc.list	19
bc.raw	20
bc.rel	21
bc.str	22
bcapply	23
bc_dim	24
bc_ifelse	25
bc_strrep	26
bind_array	27
broadcaster	30
cast_dim2flat	32
cast_dim2hier	33
cast_hier2dim	34
cast_shallow2atomic	38
dropnests	40
hier2dim	42
linear_algebra_stats	45
ndim	47
rep_dim	48
typecast	49
Index	51

aaa00_broadcast_help *broadcast Package Overview*

Description

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Broadcasted Array Operations Like 'NumPy'

Implements efficient 'NumPy'-like broadcasted operations for atomic and recursive arrays.

In the context of operations involving 2 (or more) arrays, “broadcasting” refers to efficiently recycling array dimensions without allocating additional memory.

Besides linking to 'Rcpp', 'broadcast' does not use any external libraries in any way; 'broadcast' was essentially made from scratch and can be installed out-of-the-box.

The implementations available in 'broadcast' include, but are not limited to, the following:

1. Broadcasted element-wise operations on any 2 arrays; they support a large set of relational, arithmetic, Boolean, string, and bit-wise operations.
2. A faster, more memory efficient, and broadcasted abind-like function, for binding arrays along an arbitrary dimension.
3. Broadcasted ifelse-like, and apply-like functions.
4. Casting functions, that cast subset-groups of an array to a new dimension, cast nested lists to dimensional lists, and vice-versa.
5. A few linear algebra functions for statistics.

The functions in the 'broadcast' package strive to minimize computation time and memory usage (which is not just better for efficient computing, but also for the environment).

Links to Get Started

- The Quick-Start Guide, Vignettes, Benchmarks, and more can be found on the [website](#).
- GitHub main page: <https://github.com/tony-aw/broadcast>
- Reporting Issues or Giving Suggestions: <https://github.com/tony-aw/broadcast/issues>

Functions

Broadcasted Operators

Base 'R' comes with relational (`==`, `!=`, etc.), arithmetic (`+`, `-`, `*`, `/`, etc.), and logical/bit-wise (`&`, `|`) operators.

'broadcast' provides 2 ways to use these operators with broadcasting.

The first (and simple) way is to use the [broadcaster](#) class, which comes with it's own method dispatch for the above mentioned operators.

This method support operator precedence, and for the average 'R' user, this is sufficient.

The second way is to use the large set of `bc. -` functions.

These offer much greater control and more operators than the previous method, and has less risk of running into conflicting methods.

But it does not support operator precedence.

More information about both methods can be found here:

[broadcast_operators](#).

Binding Arrays

'broadcast' provides the [bind_array](#) function, to bind arrays along an arbitrary dimension, with support for broadcasting.

See [bind_array](#).

Casting Functions

'broadcast' provides several "casting" functions.

These can facility complex forms of broadcasting that would normally not be possible.

But these "casting" functions also have their own merit, beside empowering complex broadcasting.

More information about the casting functions can be found here:
[broadcast_casting](#).

General Pairwise Broadcasted Functions

'broadcast' also comes with a few general pairwise broadcasted functions:

- [bc_ifelse](#): Broadcasted version of [ifelse](#).
- [bcapply](#): Broadcasted apply-like function.
- [bc_strrep](#): Broadcasted version of [strrep](#).

Other functions

'broadcast' provides [type-casting](#) functions, which preserve names and dimensions - convenient for arrays.

'broadcast' also provides [simple linear algebra functions for statistics](#).

And 'broadcast' comes with some helper functions:
[bc_dim](#), [ndim](#), [lst.ndim](#), [rep_dim](#).

Supported Structures

'broadcast' supports atomic/recursive arrays (up to 16 dimensions), and atomic/recursive vectors. As in standard Linear Algebra convention, dimensionless vectors are interpreted as column-vectors in broadcasted array operations.

Author(s)

Author, Maintainer: Tony Wilkes <tony_a_wilkes@outlook.com> ([ORCID](#))

References

Harris, C.R., Millman, K.J., van der Walt, S.J. et al. *Array programming with NumPy*. Nature 585, 357–362 (2020). doi:[10.1038/s4158602026492](https://doi.org/10.1038/s4158602026492). ([Publisher link](#)).

Description

Base 'R' comes with relational (`==`, `!=`, etc.), arithmetic (`+`, `-`, `*`, `/`, etc.), and logical/bit-wise (`&`, `|`) operators.

'broadcast' provides 2 ways to use these operators with broadcasting.

The first (and simple) way is to use the `broadcaster` class, which comes with its own method dispatch for the above mentioned operators.

This approach supports operator precedence, and for the average 'R' user, this is sufficient.

The second way is to use the large set of `bc.` - functions.

These offer much greater control and more operators than the previous method, and has less risk of running into conflicting methods.

But it does not support operator precedence.

Operators Overloaded via Broadcaster Class

The 'broadcast' package provides the `broadcaster` class, which comes with its own method dispatch for the base operators.

If at least one of the 2 arguments of the base operators has the `broadcaster` class attribute, and no other class (like `bit64`) interferes, broadcasting will be used.

The following arithmetic operators have a 'broadcaster' method: `+`, `-`, `*`, `/`, `^`, `%%`, `%/%`

The following relational operators have a 'broadcaster' method: `==`, `!=`, `<`, `>`, `<=`, `>=`

And finally, the `&` and `|` operators also have a 'broadcaster' method.

As the `broadcaster` operator methods simply overload the base operators, operator precedence rules are preserved for the `broadcaster` operator methods.

See also the Examples section below.

Available `bc.` functions

'broadcast' provides a set of functions for broadcasted element-wise binary operations with broadcasting.

These functions use an API similar to the `outer` function.

The following functions for simple operations are available:

- `bc.rel`: General relational operations.
- `bc.b`: Boolean (i.e. logical) operations;
- `bc.i`: integer arithmetic operations;
- `bc.d`: decimal arithmetic operations;
- `bc.cplx`: complex arithmetic operations;
- `bc.str`: string (in)equality, concatenation, and distance operations;
- `bc.raw`: operations that take in arrays of type `raw` and return an array of type `raw`;
- `bc.bit`: BIT-WISE operations, supporting the `raw` and integer types;

- `bc.list`: apply any 'R' function to 2 recursive arrays with broadcasting.

Note that the `bc.rel` method is the primary method for relational operators (`==`, `!=`, `<`, `>`, `<=`, `>=`), and provides what most user usually need in relational operators.

The various other `bc.` methods have specialized relational operators available for very specialised needs.

Attribute Handling

The `bc.` functions and the overloaded operators generally do **not** preserve attributes, unlike the base 'R' operators, except for (dim)names and the `broadcaster` class attribute (and related attributes).

Broadcasting often results in an object with more dimensions, larger dimensions, and/or larger length than the original objects.

This is relevant as some class-specific attributes are only appropriate for certain dimensions or lengths.

Custom matrix classes, for example, presumes an object to have exactly 2 dimensions.

And the various classes provided by the 'bit' package have length-related attributes.

So class attributes cannot be guaranteed to hold for the resulting objects when broadcasting is involved.

The `bc.` functions and the overloaded operators **always** preserve the "broadcaster" attribute, as this is necessary to chain together broadcasted operations.

Almost all functions provided by 'broadcast' are S3 or S4 generics;

methods can be written for them for specific classes, so that class-specific attributes can be supported when needed.

Unary operations (i.e. `+ x`, `- x`) return the original object, with only the sign adjusted.

Examples

```
# maths ====

x <- 1:10
y <- 1:10
dim(x) <- c(10, 1)
dim(y) <- c(1, 10)
broadcaster(x) <- TRUE
broadcaster(y) <- TRUE
```

```
x + y / x
(x + y) / x

(x + y) * x
```

```
# relational operators ====
x <- 1:10
y <- array(1:10, c(1, 10))
broadcaster(x) <- TRUE
broadcaster(y) <- TRUE

x == y
x != y
x < y
x > y
x <= y
x >= y
```

aaa02_broadcast_casting

Details on Casting Functions

Description

'broadcast' provides several "casting" functions.

These can facilitate complex forms of broadcasting that would normally not be possible.

But these "casting" functions also have their own merit, beside empowering complex broadcasting.

The following casting functions are available:

- [acast](#):
Casts group-based subsets of an array into a new dimension.
Useful for, for example, performing **grouped** broadcasted operations.
- [cast_hier2dim](#):
Casts a nested/hierarchical list into a dimensional list (i.e. array of type `list`).
Useful because one cannot broadcast through nesting, but one **can** broadcast along dimensions.
- [hier2dim](#), [hiernames2dimnames](#):
Helper functions for [cast_hier2dim](#).
- [cast_dim2hier](#):
Casts a dimensional list into a nested/hierarchical list; the opposite of [cast_hier2dim](#).
- [cast_shallow2atomic](#):
Casts a (dimensional) shallow (i.e. non-nested) list into an atomic vector or array.
Useful because atomic vectors/arrays have access to many vectorized (broadcasted) operations that may not be available for vectors/arrays of type `list`.
- [cast_dim2flat](#):
Casts a dimensional list into a flattened list, but with names that indicate their original dimensional positions.
Mostly useful for printing or summarizing dimensional lists.
- [dropnests](#):
Drop redundant nesting in lists; mostly used for facilitating the above casting functions.

Shared argument `recurse_all`

The `dropnests`, `hier2dim`, `hiernames2dimnames`, and `cast_hier2dim` methods all have the `recurse_all` argument.

By default `recurse_all = FALSE`, meaning these methods do not recurse through dimensional or classed lists (like `data.frames`).

Setting `recurse_all = TRUE` allows these methods to recurse through all list objects, even if they are dimensional and/or classed.

Shared Argument `in2out`

The `hier2dim`, `hiernames2dimnames`, `cast_hier2dim`, and `cast_dim2hier` methods all have the `in2out` argument.

`in2out: TRUE;`

By default `in2out` is `TRUE`.

This means the call

```
y <- cast_hier2dim(x)
```

will cast the elements of the deepest valid depth of `x` to the rows of `y`, and elements of the depth above that to the columns of `y`, and so on until the surface-level elements of `x` are cast to the last dimension of `y`.

Similarly, the call

```
x <- cast_dim2hier(y)
```

will cast the rows of `y` to the inner most elements of `x`, and the columns of `y` to one depth above that, and so on until the last dimension of `y` is cast to the surface-level elements of `x`.

Consider the nested list `x` with a depth of 3, and the recursive array `y` with 3 dimensions, where their relationship can be described as the following code:

```
y <- cast_hier2dim(x)
```

```
x <- cast_dim2hier(y).
```

Then it holds that:

`x[[i]][[j]][[k]]` corresponds to `y[[k, j, i]]`,

$\forall(i, j, k)$, provided `x[[i]][[j]][[k]]` exists.

`in2out: FALSE;`

If `in2out = FALSE`, the call

```
y <- cast_hier2dim(x, in2out = FALSE)
```

will cast the surface-level elements of `x` to the rows of `y`, and elements of the depth below that to the columns of `y`, and so on until the elements of the deepest valid depth of `x` are cast to the last dimension of `y`.

Similarly, the call

```
x <- cast_dim2hier(y, in2out = FALSE)
```

will cast the rows of `y` to the surface-level elements of `x`, and the columns of `y` to one depth below that, and so on until the last dimension of `y` is cast to the inner most elements of `x`.

Consider the nested list `x` with a depth of 3, and the recursive array `y` with 3 dimensions, where their relationship can be described with the following code:


```

y <- cast_hier2dim(x, in2out = FALSE)
x <- cast_dim2hier(y, in2out = FALSE).
Then it holds that :
x[[i]][[j]][[k]] corresponds to y[[i, j, k]],
 $\forall (i, j, k)$ , provided  $x[[i]][[j]][[k]]$  exists.

```

Examples

```

# recurse_all demonstration ====
x <- list(
  a = list(list(list(list(1:10)))),
  b = data.frame(month.abb, month.name),
  c = data.frame(month.abb),
  d = array(list(1), c(1,1,1))
)

dropnests(x) # by default, recurse_all = FALSE

dropnests(x, recurse_all = TRUE) # recurse_all = TRUE

# in2out demonstration ====
x <- list(
  group1 = list(
    class1 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    ),
    class2 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    )
  ),
  group2 = list(
    class1 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    ),
    class2 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    )
  )
)

# in2out = TRUE (default):
x2 <- cast_hier2dim(x)
dimnames(x2) <- hiernames2dimnames(x)
print(x2)
cast_dim2flat(x2[1,1,,drop = FALSE])

```

```
# in2out = FALSE:
x2 <- cast_hier2dim(x, in2out = FALSE)
dimnames(x2) <- hiernames2dimnames(x, in2out = FALSE)
print(x2)
cast_dim2flat(x2[1,1,,drop = FALSE])
```

acast

Simple and Fast Casting/Pivoting of an Array

Description

The `acast()` function spreads subsets of an array margin over a new dimension. Written in 'C' and 'C++' for high speed and memory efficiency.

Roughly speaking, `acast()` can be thought of as the "array" analogy to `data.table::dcast()`. But note 2 important differences:

- `acast()` works on arrays instead of data.tables.
- `acast()` casts into a completely new dimension (namely `ndim(x) + 1`), instead of casting into new columns.

Usage

```
acast(x, ...)
```

```
## Default S3 method:
```

```
acast(x, margin, grp, fill = FALSE, fill_val, ...)
```

Arguments

<code>x</code>	an atomic or recursive array.
<code>...</code>	further arguments passed to or from methods.
<code>margin</code>	a scalar integer, specifying the margin to cast from.
<code>grp</code>	a factor, where <code>length(grp) == dim(x)[margin]</code> , with at least 2 unique values, specifying which indices of <code>dim(x)[margin]</code> belong to which group. Each group will be cast onto a separate index of dimension <code>ndim(x) + 1</code> . Unused levels of <code>grp</code> will be dropped. Any NA values or levels found in <code>grp</code> will result in an error.
<code>fill</code>	Boolean. When factor <code>grp</code> is unbalanced (i.e. has unequally sized groups) the result will be an array where some slices have missing values, which need to be filled. If <code>fill = TRUE</code> , an unbalanced <code>grp</code> factor is allowed, and missing values will be filled with <code>fill_val</code> . If <code>fill = FALSE</code> (default), an unbalanced <code>grp</code> factor is not allowed, and providing an unbalanced factor for <code>grp</code> produces an error.
<code>fill_val</code>	scalar of the same type of <code>x</code> , giving value to use to fill in the gaps when <code>fill = TRUE</code> . The <code>fill_val</code> argument is ignored when <code>fill = FALSE</code> . If <code>fill_val</code> is missing, it is specified as follows:

- If `x` is of type `raw` and `fill = TRUE`, `fill_val` is not allowed to be missing, and an error is returned;
- If `x` is atomic but not `raw`, `fill_val` is set to `NA`;
- If `x` is of type `list`, `fill_val` is set to `list(NULL)`.

Details

For the sake of illustration, consider a matrix `x` and a grouping factor `grp`. Let the integer scalar `k` represent a group in `grp`, such that $k \in 1:nlevels(grp)$. Then the code

```
out <- acast(x, margin = 1, grp = grp)
```

essentially performs the following for every group `k`:

- copy-paste the subset `x[grp == k, ]` to the subset `out[, , k]`.

Please see the examples section to get a good idea on how this function casts an array.

Value

An array with the following properties:

- the number of dimensions of the output array is equal to `ndim(x) + 1`;
- the dimensions of the output array is equal to `c(dim(x), max(tabulate(grp)))`;
- the `dimnames` of the output array is equal to `c(dimnames(x), list(levels(grp)))`.

Back transformation

From the casted array,

```
out <- acast(x, margin, grp),
```

one can get the original `x` back by using

```
back <- asplit(out, ndim(out)) |> bind_array(along = margin).
```

Note, however, the following about the back-transformed array `back`:

- `back` will be ordered by `grp` along dimension `margin`;
- if the levels of `grp` did not have equal frequencies, then `dim(back)[margin] > dim(x)[margin]`, and `back` will have more missing values than `x`.

See Also

[broadcast_casting](#)

Examples

```
x <- cbind(id = c(rep(1:3, each = 2), 1), grp = c(rep(1:2, 3), 2), val = rnorm(7))
print(x)

grp <- as.factor(x[, 2])
levels(grp) <- c("a", "b")
margin <- 1L

acast(x, margin, grp, fill = TRUE)
```

bc.b

Broadcasted Boolean Operations

Description

The `bc.b()` function performs broadcasted logical (or Boolean) operations on 2 arrays.

Please note that these operations will treat the input as `logical`.

Therefore, something like `bc.b(1, 2, "==")` returns `TRUE`, because both 1 and 2 are `TRUE` when treated as `logical`.

For regular relational operators, see [bc.rel](#).

Usage

```
bc.b(x, y, op, ...)
```

S4 method for signature 'ANY'

```
bc.b(x, y, op)
```

Arguments

<code>x, y</code>	conformable vectors/arrays of type <code>logical</code> , <code>numeric</code> , or <code>raw</code> .
<code>op</code>	a single string, giving the operator. Supported Boolean operators: <code>&</code> , <code> </code> , <code>xor</code> , <code>nand</code> , <code>==</code> , <code>!=</code> , <code><</code> , <code>></code> , <code><=</code> , <code>>=</code> .
<code>...</code>	further arguments passed to or from methods.

Details

`bc.b()` efficiently casts the input to `logical`.

Since the input is treated as `logical`, the following equalities hold for `bc.b()`:

- `"=="` is equivalent to `(x & y) | (!x & !y)`, but faster;
- `"!="` is equivalent to `xor(x, y)`;
- `"<"` is equivalent to `(!x & y)`, but faster;
- `">"` is equivalent to `(x & !y)`, but faster;

- "`<=`" is equivalent to `(!x & y) | (y == x)`, but faster;
- "`>=`" is equivalent to `(x & !y) | (y == x)`, but faster.

Value

Normally:

A logical array/vector as a result of the broadcasted Boolean operation.

If both `x` and `y` are type of `raw`:

A raw array/vector as a result of the broadcasted Boolean operation, where `01` codes for `TRUE` and `00` codes for `FALSE`.

This is convenient as `raw` requires less memory space than `logical`.

See Also

[broadcast_operators](#)

Examples

```
x.dim <- c(4:2)
x.len <- prod(x.dim)
x.data <- sample(c(TRUE, FALSE, NA), x.len, TRUE)
x <- array(x.data, x.dim)
y <- array(1:50, c(4,1,1))

bc.b(x, y, "&")
bc.b(x, y, "|")
bc.b(x, y, "xor")
bc.b(x, y, "nand")
bc.b(x, y, "==")
bc.b(x, y, "!=")
```

bc.bit

Broadcasted Bit-wise Operations

Description

The `bc.bit()` function performs broadcasted bit-wise operations on pairs of arrays, where both arrays are of type `raw` or both arrays are of type `integer`.

Usage

```
bc.bit(x, y, op, ...)
```

```
## S4 method for signature 'ANY'
```

```
bc.bit(x, y, op)
```

Arguments

<code>x, y</code>	conformable raw or integer (32 bit) vectors/arrays.
<code>op</code>	a single string, giving the operator. Supported bit-wise operators: <code>&</code> , <code> </code> , <code>xor</code> , <code>nand</code> , <code><<</code> , <code>>></code> , <code>==</code> , <code>!=</code> , <code><</code> , <code>></code> , <code><=</code> , <code>>=</code> .
<code>...</code>	further arguments passed to or from methods.

Details

The `"&"`, `"|"`, `"xor"`, and `"nand"` operators given in `bc.bit()` perform BIT-WISE AND, OR, XOR, and NAND operations, respectively.

The relational operators given in `bc.bit()` perform BIT-WISE relational operations:

- `"=="` is equivalent to bit-wise $(x \& y) \mid (!x \& !y)$, but faster;
- `"!="` is equivalent to bit-wise `xor(x, y)`;
- `"<"` is equivalent to bit-wise $(!x \& y)$, but faster;
- `">"` is equivalent to bit-wise $(x \& !y)$, but faster;
- `"<="` is equivalent to bit-wise $(!x \& y) \mid (y == x)$, but faster;
- `">="` is equivalent to bit-wise $(x \& !y) \mid (y == x)$, but faster.

The `"<<"` and `">>"` operators perform bit-wise left-shift and right-shift, respectively, on `x` by unit `y`. For these shift operations, `y` being larger than the number of bits of `x` results in an error. Shift operations are only supported for type of `raw`.

Value

For bit-wise operators:

An array of the same type as `x` and `y`, as a result of the broadcasted bit-wise operation.

See Also

[broadcast_operators](#)

Examples

```
x.dim <- c(4:2)
x.len <- prod(x.dim)
x.data <- as.raw(0:10)
y.data <- as.raw(10:0)
x <- array(x.data, x.dim)
y <- array(y.data, c(4,1,1))

bc.bit(x, y, "&")
```

```
bc.bit(x, y, "|")
bc.bit(x, y, "xor")
```

bc.cplx

Broadcasted Complex Numeric Operations

Description

The `bc.cplx()` function performs broadcasted complex numeric operations on pairs of arrays.

Note that `bc.cplx()` uses more strict NA checks than base 'R':

If for an element of either `x` or `y`, either the real or imaginary part is NA or NaN, then the result of the operation for that element is necessarily NA.

Usage

```
bc.cplx(x, y, op, ...)
```

```
## S4 method for signature 'ANY'
```

```
bc.cplx(x, y, op)
```

Arguments

<code>x, y</code>	conformable vectors/arrays of type complex.
<code>op</code>	a single string, giving the operator. Supported arithmetic operators: <code>+</code> , <code>-</code> , <code>*</code> , <code>/</code> . Supported relational operators: <code>==</code> , <code>!=</code> .
<code>...</code>	further arguments passed to or from methods.

Value

For arithmetic operators:

A complex array as a result of the broadcasted arithmetic operation.

For relational operators:

A logical array as a result of the broadcasted relational comparison.

See Also

[broadcast_operators](#)

Examples

```
x.dim <- c(4:2)
x.len <- prod(x.dim)
gen <- function() sample(c(rnorm(10), NA, NA, NaN, NaN, Inf, Inf, -Inf, -Inf))
x <- array(gen() + gen() * -1i, x.dim)
y <- array(gen() + gen() * -1i, c(4,1,1))

bc.cplx(x, y, "==")
bc.cplx(x, y, "!=")

bc.cplx(x, y, "+")

bc.cplx(array(gen() + gen() * -1i), array(gen() + gen() * -1i), "==")
bc.cplx(array(gen() + gen() * -1i), array(gen() + gen() * -1i), "!=")

x <- gen() + gen() * -1i
y <- gen() + gen() * -1i
out <- bc.cplx(array(x), array(y), "*")
cbind(x, y, x*y, out)
```

bc.d

Broadcasted Decimal Numeric Operations

Description

The `bc.d()` function performs broadcasted decimal numeric operations on 2 numeric or logical arrays.

Usage

```
bc.d(x, y, op, ...)

## S4 method for signature 'ANY'
bc.d(x, y, op, tol = sqrt(.Machine$double.eps))
```

Arguments

<code>x, y</code>	conformable vectors/arrays of type logical or numeric.
<code>op</code>	a single string, giving the operator. Supported arithmetic operators: <code>+</code> , <code>-</code> , <code>*</code> , <code>/</code> , <code>^</code> , <code>pmin</code> , <code>pmax</code> . Supported relational operators: <code>==</code> , <code>!=</code> , <code><</code> , <code>></code> , <code><=</code> , <code>>=</code> , <code>d==</code> , <code>d!=</code> , <code>d<</code> , <code>d></code> , <code>d<=</code> , <code>d>=</code> .
<code>...</code>	further arguments passed to or from methods.
<code>tol</code>	a single number between 0 and 0.1, giving the machine tolerance to use for the relational operators. Only relevant for the following operators:

`d==, d!=, d<, d>, d<=, d>=`

See the `%d==%, %d!=%, %d<%, %d>%, %d<=%, %d>=%` operators from the 'tinycodet' package for details.

Value

For arithmetic operators:

A numeric array as a result of the broadcasted decimal arithmetic operation.

For relational operators:

A logical array as a result of the broadcasted decimal relational comparison.

See Also

[broadcast_operators](#)

Examples

```
x.dim <- c(4:2)
x.len <- prod(x.dim)
x.data <- sample(c(NA, 1.1:1000.1), x.len, TRUE)
x <- array(x.data, x.dim)
y <- array(1:50, c(4,1,1))
```

```
bc.d(x, y, "+")
bc.d(x, y, "-")
bc.d(x, y, "*")
bc.d(x, y, "/")
bc.d(x, y, "^")
```

```
bc.d(x, y, "==")
bc.d(x, y, "!=")
bc.d(x, y, "<")
bc.d(x, y, ">")
bc.d(x, y, "<=")
bc.d(x, y, ">=")
```

bc.i

Broadcasted Integer Numeric Operations with Extra Overflow Protection

Description

The `bc.i()` function performs broadcasted integer numeric operations on 2 numeric or logical arrays.

Please note that these operations will treat the input as (double typed) integers, and will efficiently truncate when necessary.

Therefore, something like `bc.i(1, 1.5, "==")` returns TRUE, because `trunc(1.5)` equals 1.

For regular relational operators, see [bc.rel](#).

Usage

```
bc.i(x, y, op, ...)
```

```
## S4 method for signature 'ANY'
```

```
bc.i(x, y, op)
```

Arguments

<code>x, y</code>	conformable vectors/arrays of type logical or numeric.
<code>op</code>	a single string, giving the operator. Supported simple arithmetic operators: <code>+</code> , <code>-</code> , <code>*</code> , <code>^</code> , <code>pmin</code> , <code>pmax</code> . Supported special division arithmetic operators: <code>gcd</code> , <code>%%</code> , <code>%/%</code> . Supported relational operators: <code>==</code> , <code>!=</code> , <code><</code> , <code>></code> , <code><=</code> , <code>>=</code> . The "gcd" operator performs the "Greatest Common Divisor" operation, using the Euclidean algorithm.
<code>...</code>	further arguments passed to or from methods.

Value

For arithmetic operators:

A numeric array of whole numbers, as a result of the broadcasted arithmetic operation.

Base 'R' supports integers from -2^{53} to 2^{53} , which thus range from approximately -9 quadrillion to +9 quadrillion.

Values outside of this range will be returned as `-Inf` or `Inf`, as an extra protection against integer overflow.

For relational operators:

A logical array as a result of the broadcasted integer relational comparison.

See Also

[broadcast_operators](#)

Examples

```
x.dim <- c(4:2)
x.len <- prod(x.dim)
x.data <- sample(c(NA, 1.1:1000.1), x.len, TRUE)
x <- array(x.data, x.dim)
y <- array(1:50, c(4,1,1))

bc.i(x, y, "+")
bc.i(x, y, "-")
bc.i(x, y, "*")
```

```

bc.i(x, y, "gcd") # greatest common divisor
bc.i(x, y, "^")

bc.i(x, y, "==")
bc.i(x, y, "!=")
bc.i(x, y, "<")
bc.i(x, y, ">")
bc.i(x, y, "<=")
bc.i(x, y, ">=")

```

bc.list

Broadcasted Operations for Recursive Arrays

Description

The `bc.list()` function performs broadcasted operations on 2 Recursive arrays.

Usage

```

bc.list(x, y, f, ...)

## S4 method for signature 'ANY'
bc.list(x, y, f)

```

Arguments

<code>x, y</code>	conformable Recursive vectors/arrays (i.e. vectors/arrays of type <code>list</code>).
<code>f</code>	a function that takes in exactly 2 arguments, and returns a result that can be stored in a single element of a list.
<code>...</code>	further arguments passed to or from methods.

Value

A recursive array.

See Also

[broadcast_operators](#)

Examples

```

x.dim <- c(10, 2, 2)
x.len <- prod(x.dim)

gen <- function(n) sample(list(letters, month.abb, 1:10), n, TRUE)

```

```
x <- array(gen(10), x.dim)
y <- array(gen(10), c(10,1,1))

bc.list(
  x, y,
  \ (x, y) c(length(x) == length(y), typeof(x) == typeof(y))
)
```

bc.raw

Broadcasted Operations that Take Raw Arrays and Return Raw Arrays

Description

The `bc.raw()` function performs broadcasted operations on arrays of type `raw`, and the return type is **always** `raw`.

For bit-wise operations, use [bc.bit](#).

For relational operations with logical (TRUE/FALSE/NA) results, use [bc.rel](#).

Usage

```
bc.raw(x, y, op, ...)
```

```
## S4 method for signature 'ANY'
bc.raw(x, y, op)
```

Arguments

<code>x, y</code>	conformable vectors/arrays of type <code>raw</code> .
<code>op</code>	a single string, giving the operator. Supported operators: <code>==</code> , <code>!=</code> , <code><</code> , <code>></code> , <code><=</code> , <code>>=</code> , <code>pmin</code> , <code>pmax</code> , <code>diff</code> . The relational operators work the same as in bc.rel , but with the following difference: a TRUE result is replaced with <code>01</code> , and a FALSE result is replaced with <code>00</code> . The "diff" operator performs the byte equivalent of <code>abs(x - y)</code> .
<code>...</code>	further arguments passed to or from methods.

Value

`bc.raw()` **always** returns an array of type `raw`.

For the relational operators, `01` codes for TRUE results, and `00` codes for FALSE results.

See Also

[broadcast_operators](#)

Examples

```
x <- array(
  sample(as.raw(1:100)), c(5, 3, 2)
)
y <- array(
  sample(as.raw(1:100)), c(5, 1, 1)
)

cond <- bc.raw(x, y, "!=")
print(cond)

bc_ifelse(cond, yes = x, no = y)
```

bc.rel

*Broadcasted General Relational Operators***Description**

The `bc.rel()` function performs broadcasted general relational operations on 2 arrays.

Usage

```
bc.rel(x, y, op, ...)

## S4 method for signature 'ANY'
bc.rel(x, y, op)
```

Arguments

<code>x, y</code>	conformable vectors/arrays of any atomic type.
<code>op</code>	a single string, giving the relational operator. Supported relational operators: <code>==</code> , <code>!=</code> , <code><</code> , <code>></code> , <code><=</code> , <code>>=</code> .
<code>...</code>	further arguments passed to or from methods.

Value

A logical array as a result of the broadcasted general relational operation.

See Also

[broadcast_operators](#)

Examples

```
x.dim <- c(4:2)
x.len <- prod(x.dim)
x.data <- sample(c(NA, 1.1:1000.1), x.len, TRUE)
x <- array(x.data, x.dim)
y <- array(1:50, c(4,1,1))
```

```
bc.rel(x, y, "==")
bc.rel(x, y, "!=")
bc.rel(x, y, "<")
bc.rel(x, y, ">")
bc.rel(x, y, "<=")
bc.rel(x, y, ">=")
```

bc.str

*Broadcasted String Operations***Description**

The `bc.str()` function performs broadcasted string operations on pairs of arrays.

Usage

```
bc.str(x, y, op, ...)
```

S4 method for signature 'ANY'

```
bc.str(x, y, op)
```

Arguments

<code>x, y</code>	conformable vectors/arrays of type character.
<code>op</code>	a single string, giving the operator. Supported concatenation operators: +. Supported relational operators: ==, !=. Supported distance operators: levenshtein.
<code>...</code>	further arguments passed to or from methods.

Value

For concatenation operation:
A character array as a result of the broadcasted concatenation operation.

For relational operation:
A logical array as a result of the broadcasted relational comparison.

For distance operation:

An integer array as a result of the broadcasted distance measurement.

References

The 'C++' code for the Levenshtein edit string distance is based on the code found in https://rosettacode.org/wiki/Levenshtein_distance#C++

See Also

[broadcast_operators](#)

Examples

```
# string concatenation:
x <- array(letters, c(10, 2, 1))
y <- array(letters, c(10,1,1))
bc.str(x, y, "+")

# string (in)equality:
bc.str(array(letters), array(letters), "==")
bc.str(array(letters), array(letters), "!=")

# string distance (Levenshtein):
x <- array(month.name, c(12, 1))
y <- array(month.abb, c(1, 12))
out <- bc.str(x, y, "levenshtein")
dimnames(out) <- list(month.name, month.abb)
print(out)
```

bcapply

Apply Function to Pair of Arrays with Broadcasting

Description

The `bcapply()` method applies a function to 2 arrays element-wise with broadcasting.

Usage

```
bcapply(x, y, f, ...)
```

S4 method for signature 'ANY'

```
bcapply(x, y, f, v = NULL)
```

Arguments

<code>x, y</code>	conformable atomic or recursive vectors/arrays.
<code>f</code>	a function that takes in exactly 2 arguments, and returns a result that can be stored in a single element of a recursive or atomic array.
<code>...</code>	further arguments passed to or from methods.
<code>v</code>	either NULL, or single string, giving the scalar type for a single iteration. If NULL (default) or "list", the result will be a recursive array. If it is certain that, for every iteration, <code>f()</code> always results in a single atomic scalar , the user can specify the type in <code>v</code> to pre-allocate the result. Pre-allocating the results leads to slightly faster and more memory efficient code. NOTE: Incorrectly specifying <code>v</code> leads to undefined behaviour; when unsure, leave <code>v</code> at its default value.

Value

An atomic or recursive array with dimensions `bc_dim(x, y)`.
Preserves some of the attributes of `x` and `y` similar to broadcasted infix operators, as explained in [broadcast_operators](#).

Examples

```
x.dim <- c(5, 3, 2)
x.len <- prod(x.dim)

gen <- function(n) sample(list(letters, month.abb, 1:10), n, TRUE)

x <- array(gen(10), x.dim)
y <- array(1:5, c(5, 1, 1))

f <- function(x, y) strrep(x, y)
bcapply(x, y, f, v = "character")
```

`bc_dim`*Predict Broadcasted Dimensions*

Description

`bc_dim(x, y)` gives the dimensions an array would have, as the result of an broadcasted binary element-wise operation between 2 arrays `x` and `y`.

Usage

```
bc_dim(x, y)
```

Arguments

<code>x, y</code>	an atomic or recursive array.
-------------------	-------------------------------

Value

Returns an integer vector giving the broadcasted dimension sizes of the result, or the length of the result if its dimensions will be NULL.

Examples

```
x.dim <- c(4:2)
x.len <- prod(x.dim)
x.data <- sample(c(TRUE, FALSE, NA), x.len, TRUE)
x <- array(x.data, x.dim)
y <- array(1:50, c(4,1,1))

dim(bc.b(x, y, "&")) == bc_dim(x, y)
dim(bc.b(x, y, "|")) == bc_dim(x, y)
```

bc_ifelse

*Broadcasted Ifelse***Description**

The `bc_ifelse()` method performs a broadcasted form of [ifelse](#).

Usage

```
bc_ifelse(test, yes, no, ...)

## S4 method for signature 'ANY'
bc_ifelse(test, yes, no)
```

Arguments

test	a vector or array, with the type logical, integer, or raw, and a length equal to <code>prod(bc_dim(yes, no))</code> . If yes / no are of type raw, test is not allowed to contain any NAs.
yes, no	conformable vectors/arrays of the same type. All atomic types are supported. Recursive arrays of type list are also supported.
...	further arguments passed to or from methods.

Value

The output, here referred to as out, will be an array of the same type as yes and no.
If test has the same dimensions as `bc_dim(yes, no)`, then out will also have the same dimnames as test.
If test is a [broadcaster](#), then out will also be a [broadcaster](#).

After broadcasting yes against no, given any element index *i*, the following will hold for the output:

- when `test[i] == TRUE`, `out[i]` is `yes[i]`;
- when `test[i] == FALSE`, `out[i]` is `no[i]`;
- when `test[i]` is NA, `out[i]` is NA when yes and no are atomic, and `out[i]` is `list(NULL)` when yes and no are recursive.

Examples

```
x.dim <- c(5, 3, 2)
x.len <- prod(x.dim)

x <- array(sample(1:100), x.dim)
y <- array(sample(1:100), c(5, 1, 1))

cond <- bc.i(x, y, ">")

bc_ifelse(cond, yes = x^2, no = -y)
```

bc_strep

Broadcasted strep

Description

The `bc_strep()` method is a broadcasted form of [strep](#).

Usage

```
bc_strep(x, y, ...)

## S4 method for signature 'ANY'
bc_strep(x, y)
```

Arguments

<code>x</code>	vector/array of type character.
<code>y</code>	vector/array of type integer.
<code>...</code>	further arguments passed to or from methods.

Value

A character array as a result of the broadcasted repetition operation.

Examples

```
x <- array(sample(month.abb), c(10, 2))
y <- array(sample(1:10), c(10, 2, 3))

print(x)
print(y)

bc_strrep(x, y)
```

bind_array

*Dimensional Binding of Arrays with Broadcasting***Description**

bind_array() binds (atomic/recursive) arrays and (atomic/recursive) matrices along a dimension. Allows for broadcasting.

Usage

```
bind_array(
  input,
  along,
  rev = FALSE,
  ndim2bc = 16L,
  name_along = TRUE,
  comnames_from = 1L
)
```

Arguments

- | | |
|-------|--|
| input | <p>a list of arrays; both atomic and recursive arrays are supported, and can be mixed.</p> <p>If argument input has length 0, or it contains exclusively objects where one or more dimensions are 0, an error is returned.</p> <p>If input has length 1, bind_array() simply returns input[[1L]].</p> <p>input may not contain more than 2¹⁶ objects.</p> |
| along | <p>a single integer, indicating the dimension along which to bind the dimensions. I.e. use along = 1 for row-binding, along = 2 for column-binding, etc.</p> <p>Specifying along = 0 will bind the arrays on a new dimension before the first, making along the new first dimension.</p> <p>Specifying along = N + 1, with N = max(1st.ndim(input)), will create an additional dimension (N + 1) and bind the arrays along that new dimension.</p> |
| rev | <p>Boolean, indicating if along should be reversed, counting backwards.</p> <p>If FALSE (default), along works like normally; if TRUE, along is reversed.</p> <p>I.e. along = 0, rev = TRUE is equivalent to along = N+1, rev = FALSE; and along = N+1, rev = TRUE is equivalent to along = 0, rev = FALSE; with N = max(1st.ndim(input)).</p> |

ndim2bc	a single non-negative integer; specify here the maximum number of dimensions that are allowed to be broadcasted when binding arrays. If ndim2bc = 0L, no broadcasting will be allowed at all.
name_along	Boolean, indicating if dimension along should be named. Please run the code in the examples section to get a demonstration of the naming behaviour.
comnames_from	either an integer scalar or NULL. Indicates which object in input should be used for naming the shared dimension. If NULL, no communal names will be given. For example: When binding columns of matrices, the matrices will share the same rownames. Using comnames_from = 10 will then result in bind_array() using rownames(input[[10]]) for the rownames of the output.

Details

The API of bind_array() is inspired by the fantastic abind::abind() function by Tony Plare & Richard Heiberger (2016).

But bind_array() differs considerably from abind::abind in the following ways:

- bind_array() allows for broadcasting, while abind::abind does not support broadcasting.
- bind_array() is generally faster and more memory-efficient than abind::abind, as bind_array() relies heavily on 'C' and 'C++' code.
- bind_array() differs from abind::abind in that it can handle recursive arrays properly (the abind::abind function would unlist everything to atomic arrays, ruining the structure).
- unlike abind::abind, bind_array() only binds (atomic/recursive) arrays and matrices. bind_array() does not attempt to convert things to arrays when they are not arrays, but will give an error instead.
This saves computation time and prevents unexpected results.
- bind_array() has more streamlined naming options, compared to abind::abind.

Value

An array as a result from the (broadcasted) binding.

The type of the result is determined from the highest type of any of the inputs.

The hierarchy of types is:

raw < logical < integer < double < complex < character < list .

References

Plate T, Heiberger R (2016). *abind: Combine Multidimensional Arrays*. R package version 1.4-5, <https://CRAN.R-project.org/package=abind>.

Examples

```

# Simple example ====
x <- array(1:20, c(5, 4))
y <- array(-1:-15, c(5, 3))
z <- array(21:40, c(5, 4))
input <- list(x, y, z)
# column binding:
bind_array(input, 2L)

# Broadcasting example ====
x <- array(1:20, c(5, 4))
y <- array(-1:-5, c(1, 5)) # rows will be broadcasted from 1 to 5
z <- array(21:40, c(5, 4))
input <- list(x, y, z)
bind_array(input, 2L)

# Mixing types ====
# here, atomic and recursive arrays are mixed,
# resulting in recursive arrays

# creating the arrays:
x <- c(
  lapply(1:3, \(x)sample(c(TRUE, FALSE, NA))),
  lapply(1:3, \(x)sample(1:10)),
  lapply(1:3, \(x)rnorm(10)),
  lapply(1:3, \(x)sample(letters))
) |> matrix(4, 3, byrow = TRUE)
dimnames(x) <- list(letters[1:4], LETTERS[1:3])
print(x)

y <- matrix(1:12, 4, 3)
print(y)
z <- matrix(letters[1:12], c(4, 3))

# column-binding:
input <- list(x = x, y = y, z = z)
bind_array(input, along = 2L)

# Illustrating `along` argument ====
# using recursive arrays for clearer visual distinction
input <- list(x = x, y = y)

bind_array(input, along = 0L) # binds on new dimension before first
bind_array(input, along = 1L) # binds on first dimension (i.e. rows)
bind_array(input, along = 2L)
bind_array(input, along = 3L) # bind on new dimension after last

bind_array(input, along = 0L, TRUE) # binds on new dimension after last
bind_array(input, along = 1L, TRUE) # binds on last dimension (i.e. columns)
bind_array(input, along = 2L, TRUE)
bind_array(input, along = 3L, TRUE) # bind on new dimension before first

```

```
# binding, with empty arrays ====
emptyarray <- array(numeric(0L), c(0L, 3L))
dimnames(emptyarray) <- list(NULL, paste("empty", 1:3))
print(emptyarray)
input <- list(x = x, y = emptyarray)
bind_array(input, along = 1L, comnames_from = 2L) # row-bind

# Illustrating `name_along` ====
x <- array(1:20, c(5, 3), list(NULL, LETTERS[1:3]))
y <- array(-1:-20, c(5, 3))
z <- array(-1:-20, c(5, 3))

bind_array(list(a = x, b = y, z), 2L)
bind_array(list(x, y, z), 2L)
bind_array(list(a = unname(x), b = y, c = z), 2L)
bind_array(list(x, a = y, b = z), 2L)
input <- list(x, y, z)
names(input) <- c("", NA, "")
bind_array(input, 2L)
```

broadcaster

Check or Set if an Array is a Broadcaster

Description

`broadcaster()` checks if an array has the "broadcaster" attribute.
`broadcaster()<-` sets or un-sets the class attribute "broadcaster" on an array.

The broadcaster class attribute exists purely to overload the arithmetic, Boolean, bit-wise, and relational infix operators, to support broadcasting.

This makes mathematical expressions with multiple variables, where precedence may be important, far more convenient.

Like in the following calculation:

$x / (y + z)$

See [broadcast_operators](#) for more information.

Usage

`broadcaster(x)`

`broadcaster(x) <- value`

Arguments

x object to check or set.
 Only S3 vectors and arrays are supported, and only up to 16 dimensions.

value set to TRUE to make an array a broadcaster, or FALSE to remove the broadcaster class attribute from an array.

Value

For `broadcaster()`:
TRUE if an array or vector is a broadcaster, or FALSE if it is not.

For `broadcaster()<=`:
Returns nothing, but sets (if right hand side is TRUE) or removes (if right hand side is FALSE) the "broadcaster" class attribute.

See Also

[broadcast_operators](#)

Examples

```
# maths ====

x <- 1:10
y <- 1:10
dim(x) <- c(10, 1)
dim(y) <- c(1, 10)
broadcaster(x) <- TRUE
broadcaster(y) <- TRUE


x + y / x
(x + y) / x

(x + y) * x


# relational operators ====
x <- 1:10
y <- array(1:10, c(1, 10))
broadcaster(x) <- TRUE
broadcaster(y) <- TRUE


x == y
x != y
x < y
x > y
x <= y
x >= y
```

cast_dim2flat

Cast Dimensional List into a Flattened List

Description

cast_dim2flat() casts a dimensional list (i.e. recursive array) into a flat list (i.e. recursive vector), but with names that indicate the original dimensional positions of the elements.

Primary purpose for this function is to facilitate printing or summarizing dimensional lists.

Usage

```
cast_dim2flat(x, ...)
```

```
## Default S3 method:
cast_dim2flat(x, ...)
```

Arguments

x a list
 ... further arguments passed to or from methods.

Value

A flattened list, with names that indicate the original dimensional positions of the elements.

See Also

[broadcast_casting](#)

Examples

```
x <- array(
  sample(list(letters, month.name, 1:10 ~ "foo"), prod(4:2), TRUE),
  dim = 4:2,
  dimnames = list(NULL, LETTERS[1:3], c("x", "y"))
)

# summarizing ====
summary(x) # dimensional information is lost

# In the following instances, dimensional position info is retained:
cast_dim2flat(x) |> summary()
```



```
cast_dim2flat(x[1:3, 1:2, 2, drop = FALSE]) |> summary()

cast_dim2flat(x[1:3, 1:2, 2, drop = TRUE]) |> summary()

# printing ====
print(x) # too compact
cast_dim2flat(x) |> print() # much less compact
```

cast_dim2hier	<i>Cast Dimensional List into Hierarchical List</i>
---------------	---

Description

cast_dim2hier() casts a dimensional list (i.e. an array of type list) into a hierarchical/nested list.

Usage

```
cast_dim2hier(x, ...)

## Default S3 method:
cast_dim2hier(x, in2out = TRUE, distr.names = TRUE, ...)
```

Arguments

x	an array of type list.
...	further arguments passed to or from methods.
in2out	see broadcast_casting .
distr.names	TRUE or FALSE, indicating if dimnames from x should be distributed over the nested elements of the output. See examples section for demonstration.

Value

A nested list.

See Also

[broadcast_casting](#)

Examples

```
x <- array(c(as.list(1:24), as.list(letters)), 4:2)
dimnames(x) <- list(
  letters[1:4],
  LETTERS[1:3],
  month.abb[1:2]
)
print(x)

# cast `x` from in to out, and distribute names:
x2 <- cast_dim2hier(x, distr.names = TRUE)
head(x2, n = 2)

# cast `x` from out to in, and distribute names:
x2 <- cast_dim2hier(x, in2out = FALSE, distr.names = TRUE)
head(x2, n = 2)
```

cast_hier2dim

Cast Hierarchical List into Dimensional list

Description

cast_hier2dim() casts a hierarchical/nested list into a dimensional list (i.e. an array of type list).

This method comes with 2 helper functions:
[hier2dim](#) and [hiernames2dimnames](#) methods.
 See their help page for details.

Usage

```
cast_hier2dim(x, ...)

## Default S3 method:
cast_hier2dim(
  x,
  in2out = TRUE,
  maxdepth = 16L,
  recurse_all = FALSE,
  padding = list(NULL),
  direction.names = 0L,
  ...
)
```

Arguments

x	a nested list. If x has redundant nesting, it is advisable (though not necessary) to reduce the redundant nesting using dropnests .
---	--

...	further arguments passed to or from methods.
in2out, recurse_all	see broadcast_casting .
maxdepth	a single, positive integer, giving the maximum depth to recurse into the list. The surface-level elements of a list is depth 1.
padding	a list of length 1, giving the padding value to use when padding is required. Padding is used to ensure every all slices of the same dimension in the output have equal number of elements (for example, all rows must have the same number of columns).
direction.names	see argument direction from the hiernames2dimnames method.

Value

An array of type list, with the dimensions given by [hier2dim](#).
If the output needs padding (indicated by [hier2dim](#)), the output will have more elements than x, filled with a padding value (as specified in the padding argument).

See Also

[broadcast_casting](#), [hier2dim](#), [hiernames2dimnames](#)

Examples

```
# Example 1: Basics ====
x <- list(
  group1 = list(
    class1 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    ),
    class2 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    )
  ),
  group2 = list(
    class1 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    ),
    class2 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
```

```

    sex = sample(c("M", "F", NA), 10, TRUE)
  )
)

# predict what dimensions `x` would have if casted as dimensional:
hier2dim(x)

x2 <- cast_hier2dim(x) # cast as dimensional

# since the original list uses the same names for all elements within the same depth,
# dimnames can be set easily:
dimnames(x2) <- hiernames2dimnames(x)

print(x2)

# Example 2: Cast from outside to inside ====
x <- list(
  group1 = list(
    class1 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    ),
    class2 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    )
  ),
  group2 = list(
    class1 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    ),
    class2 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    )
  )
)

# by default, `in2out = TRUE`;
# for this example, `in2out = FALSE` is used

# predict what dimensions `x` would have if casted as dimensional:
hier2dim(x, in2out = FALSE)

x2 <- cast_hier2dim(x, in2out = FALSE) # cast as dimensional

# since the original list uses the same names for all elements within the same depth,
# dimnames can be set easily:
# because in2out = FALSE, go from the shallow names to the deeper names:
dimnames(x2) <- hiernames2dimnames(x, in2out = FALSE)

```

```

print(x2)

# Example 3: padding ====

# For Example 3, take the same list as before, but remove x$group1$class2:

x <- list(
  group1 = list(
    class1 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    )
  ),
  group2 = list(
    class1 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    ),
    class2 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    )
  )
)

hier2dim(x) # as indicated here, dimension 2 (i.e. columns) will have padding

# casting this to a dimensional list will resulting in padding with `NULL`:
x2 <- cast_hier2dim(x)
print(x2)
# The `NULL` values are added for padding.
# This is because all slices of the same dimension need to have the same number of elements.
# For example, all rows need to have the same number of columns.

# one can also use custom padding:
x2 <- cast_hier2dim(x, padding = list(~ "this is padding"))
print(x2)

dimnames(x2) <- hiernames2dimnames(x)

print(x2)

# we can also use in2out = FALSE:
x2 <- cast_hier2dim(x, in2out = FALSE, padding = list(~ "this is padding"))
dimnames(x2) <- hiernames2dimnames(x, in2out = FALSE)
print(x2)

```

cast_shallow2atomic	<i>Cast Shallow List to Atomic Object</i>
---------------------	---

Description

cast_shallow2atomic() casts a shallow (i.e. non-nested) list to an atomic object.

Usage

```
cast_shallow2atomic(x, ...)
```

```
## Default S3 method:
```

```
cast_shallow2atomic(x, arrangement = 0L, padding = NA, comnames_from = 1L, ...)
```

Arguments

x	a shallow (i.e. non-nested) list. The attributes of the objects inside the list will be ignored, except for names.
...	further arguments passed to or from methods.
arrangement	see the Details and Examples sections.
padding	an atomic scalar, and only relevant if arrangement is 1 or -1. This gives the padding value to use when padding is required. Padding is used to ensure every all slices of the same dimension in the output have equal number of elements (for example, all rows must have the same number of columns).
comnames_from	an integer scalar or NULL, and only relevant if arrangement is 1 or -1. This gives which element of x to use for the communal names. If NULL, no communal names will be given. For example: If x is a 1d (or dimensionless) list, cast_shallow2atomic(x, 1) will produce an atomic matrix. The column names of the matrix will be names(x). The row names, however, will be taken from names(x[[comnames_from]]), provided that x[[comnames_from]] has the proper length. See also the Examples section.

Details

If arrangement = 0L,
cast_shallow2atomic() works like unlist(), except that cast_shallow2atomic() guarantees an atomic vector result.

If arrangement = 1L,
cast_shallow2atomic() will produce an atomic array, with the elements arranged such that the dimensions are c(max(lengths(x)), dim(x)).

If x has no dimensions, dim(x) is replaced with length(x), thus treating x as an 1d array.

This will therefore always produce an atomic array with at least 2 dimensions.

If arrangement = -1L,

cast_shallow2atomic() will produce an atomic array, with the elements arranged such that the dimensions are c(dim(x), max(lengths(x))).

If x has no dimensions, dim(x) is replaced with length(x), thus treating x as an 1d array.

This will therefore always produce an atomic array with at least 2 dimensions.

Value

If arrangement = 0L:

An atomic vector without dimensions.

If arrangement = 1L:

An atomic array with dimensions c(max(lengths(x)), dim(x)).

The dimnames, if possible to construct, will be c(names(x[[comnames_from]]), dimnames(x)), provided If x has no dimensions, dim(x) is replaced with length(x).

If arrangement = -1L:

An atomic array with dimensions c(dim(x), max(lengths(x))).

If x has no dimensions, dim(x) is replaced with length(x).

Back transformation

From the casted atomic object,

```
out <- cast_shallow2atomic(x, ...),
```

one can get an *approximation* of the original shallow list back using just base 'R' functions.

This section describes how to do so.

arrangement = 0L

If arrangement = 0L, one can transform an atomic object out back to a shallow list using:

```
back <- as.list(out)
```

```
names(back) <- names(out)
```

arrangement = 1L

If arrangement = 1L, one can transform an atomic object out back to a shallow list using:

```
asplit(out, seq(2, ndim(out)))
```

arrangement = -1L

If arrangement = -1L, one can transform an atomic object out back to a shallow list using:

```
asplit(out, seq(1, ndim(out) - 1L))
```

See Also[broadcast_casting](#)**Examples**

```
# recursive vector ====
x <- list(
  setNames(1:11, letters[1:11]), 1:10, 1:9, 1:8, 1:7, 1:6, 1:5, 1:4, 1:3, 1:2, 1L, integer(0L)
)
names(x) <- month.abb
print(x)

cast_shallow2atomic(x, 0L)
cast_shallow2atomic(x, 1L, comnames_from = 1L)
cast_shallow2atomic(x, -1L, comnames_from = 1L)

# recursive matrix ====
x <- list(
  setNames(1:11, letters[1:11]), 1:10, 1:9, 1:8, 1:7, 1:6, 1:5, 1:4, 1:3, 1:2, 1L, integer(0L)
) |> rev()
dim(x) <- c(3, 4)
dimnames(x) <- list(month.abb[1:3], month.name[1:4])
print(x)

cast_shallow2atomic(x, 0L)
cast_shallow2atomic(x, 1L, comnames_from = length(x))
cast_shallow2atomic(x, -1L, comnames_from = length(x))
```

dropnests

Drop Redundant List Nesting

Description

dropnests() drops redundant nesting of a list.
 It is the hierarchical equivalent to the dimensional base::drop() function.

Usage

```
dropnests(x, ...)
```

Default S3 method:

```
dropnests(x, maxdepth = 16L, recurse_all = FALSE, ...)
```

Arguments

x a list

... further arguments passed to or from methods.

maxdepth a single, positive integer, giving the maximum depth to recurse into the list.
The surface-level elements of a list is depth 1.
dropnests(x, maxdepth = 1) will return x unchanged.

recurse_all see [broadcast_casting](#).

Value

A list without redundant nesting.
Attributes are preserved.

See Also

[broadcast_casting](#)

Examples

```
x <- list(
  a = list(list(list(list(1:10)))),
  b = list(1:10)
)

print(x)

dropnests(x)

# recurse_all demonstration ====
x <- list(
  a = list(list(list(list(1:10)))),
  b = data.frame(month.abb, month.name),
  c = data.frame(month.abb),
  d = array(list(1), c(1,1,1))
)

dropnests(x) # by default, recurse_all = FALSE

dropnests(x, recurse_all = TRUE)

# maxdepth demonstration ====
x <- list(
  a = list(list(list(list(1:10)))),
  b = list(1:10)
)
print(x)

dropnests(x) # by default, maxdepth = 16

dropnests(x, maxdepth = 3L)
```

```
dropnests(x, maxdepth = 1L) # returns `x` unchanged
```

hier2dim	<i>Helper Functions For cast_hier2dim</i>
----------	---

Description

hier2dim() takes a hierarchical/nested list, and predicts what dimensions the list would have, if casted by the [cast_hier2dim](#) function.

hiernames2dimnames() takes a hierarchical/nested list, and intelligently tries to compose dimnames for the result of [cast_hier2dim](#).

Usage

```
hier2dim(x, ...)

hiernames2dimnames(x, ...)

## Default S3 method:
hier2dim(x, in2out = TRUE, maxdepth = 16L, recurse_all = FALSE, ...)

## Default S3 method:
hiernames2dimnames(
  x,
  in2out = TRUE,
  maxdepth = 16L,
  recurse_all = FALSE,
  direction = 1,
  ...
)
```

Arguments

x	a nested list. If x has redundant nesting, it is advisable (though not necessary) to reduce the redundant nesting using dropnests .
...	further arguments passed to or from methods.
in2out, recurse_all	see broadcast_casting .
maxdepth	a single, positive integer, giving the maximum depth to recurse into the list. The surface-level elements of a list is depth 1.
direction	A single number, giving the direction in which to search for names. Must be either 1 (to search from start to end) or -1 (to search from end to start). If set to 0, the result will simply be NULL.

Value

For `hier2dim()`:

An integer vector, giving the dimensions `x` would have, if casted by `cast_hier2dim()`.

The names of the output indicates if padding is required (name "padding"), or no padding is required (no name) for that dimension;

Padding will be required if not all list-elements at a certain depth have the same length.

For `hiernames2dimnames()`:

A list of `dimnames`; these can be assigned to the `dimnames` of the result of `cast_hier2dim`.

See Also

[broadcast_casting](#), [cast_hier2dim](#)

Examples

```
# Example 1: Basics ====
x <- list(
  group1 = list(
    class1 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    ),
    class2 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    )
  ),
  group2 = list(
    class1 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    ),
    class2 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    )
  )
)

# predict what dimensions `x` would have if casted as dimensional:
hier2dim(x)

x2 <- cast_hier2dim(x) # cast as dimensional

# since the original list uses the same names for all elements within the same depth,
# dimnames can be set easily:
```

```

dimnames(x2) <- hiernames2dimnames(x)

print(x2)

# Example 2: Cast from outside to inside ====
x <- list(
  group1 = list(
    class1 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    ),
    class2 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    )
  ),
  group2 = list(
    class1 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    ),
    class2 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    )
  )
)

# by default, `in2out = TRUE`;
# for this example, `in2out = FALSE` is used

# predict what dimensions `x` would have if casted as dimensional:
hier2dim(x, in2out = FALSE)

x2 <- cast_hier2dim(x, in2out = FALSE) # cast as dimensional

# since the original list uses the same names for all elements within the same depth,
# dimnames can be set easily:
# because in2out = FALSE, go from the shallow names to the deeper names:
dimnames(x2) <- hiernames2dimnames(x, in2out = FALSE)

print(x2)

# Example 3: padding ====

# For Example 3, take the same list as before, but remove x$group1$class2:
x <- list(
  group1 = list(
    class1 = list(

```

```

      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    )
  ),
  group2 = list(
    class1 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    ),
    class2 = list(
      height = rnorm(10, 170),
      weight = rnorm(10, 80),
      sex = sample(c("M", "F", NA), 10, TRUE)
    )
  )
)

```

`hier2dim(x)` # as indicated here, dimension 2 (i.e. columns) will have padding

```

# casting this to a dimensional list will resulting in padding with `NULL`:
x2 <- cast_hier2dim(x)
print(x2)
# The `NULL` values are added for padding.
# This is because all slices of the same dimension need to have the same number of elements.
# For example, all rows need to have the same number of columns.

```

```

# one can also use custom padding:
x2 <- cast_hier2dim(x, padding = list(~ "this is padding"))
print(x2)

```

```
dimnames(x2) <- hiernames2dimnames(x)
```

```
print(x2)
```

```

# we can also use in2out = FALSE:
x2 <- cast_hier2dim(x, in2out = FALSE, padding = list(~ "this is padding"))
dimnames(x2) <- hiernames2dimnames(x, in2out = FALSE)
print(x2)

```

Description

'broadcast' provides some simple Linear Algebra Functions for Statistics:

```

cinv();
sd_lc().

```

Usage

```
cinv(x)
```

```
sd_lc(X, vc, bad_rp = NaN)
```

Arguments

x	a real symmetric positive-definite square matrix.
X	a numeric (or logical) matrix of multipliers/constants
vc	the variance-covariance matrix for the (correlated) random variables.
bad_rp	if vc is not a Positive (semi-) Definite matrix, give here the value to replace bad standard deviations with.

Details

cinv()

`cinv()` computes the Choleski inverse of a real symmetric positive-definite square matrix.

sd_lc()

Given the linear combination $X \%*\% b$, where:

- X is a matrix of multipliers/constants;
- b is a vector of (correlated) random variables;
- vc is the symmetric variance-covariance matrix for b;

`sd_lc(X, vc)` computes the standard deviations for the linear combination $X \%*\% b$, without making needless copies.

`sd_lc(X, vc)` will use **much** less memory than a base 'R' approach.

`sd_lc(X, vc)` will *usually* be faster than a base 'R' approach (depending on the Linear Algebra Library used for base 'R').

Value

For `cinv()`:
A matrix.

For `sd_lc()`:
A vector of standard deviations.

References

John A. Rice (2007), *Mathematical Statistics and Data Analysis* (6th Edition)

See Also[chol](#), [chol2inv](#)**Examples**

```
vc <- datasets::ability.cov$cov
X <- matrix(rnorm(100), 100, ncol(vc))

solve(vc)
cinv(vc) # faster than `solve()`, but only works on positive definite matrices
all(round(solve(vc), 6) == round(cinv(vc), 6)) # they're the same

sd_lc(X, vc)
```

ndim*Get the Number of Dimensions of an Array*

Description

`ndim()` returns the number of dimensions of an object.
`lst.ndim()` returns the number of dimensions of every list-element.
`undim()` returns a copy of an object, but with its dimensions removed (it somewhat like the dimensional version of `unlist()`).

Usage

```
ndim(x)

lst.ndim(x)

undim(x)
```

Arguments

`x` a vector or array (for `ndim()`), or a list of vectors/arrays (for `lst.ndim()`).

Value

For `ndim()`: an integer scalar.
For `lst.ndim()`: an integer vector, with the same length, names and dimensions as `x`.
For `undim()`: the original object, but without dimensions.

Examples

```

x <- array(1:24, 2:4)
ndim(x)

x <- list(
  array(1:10, 10),
  array(1:10, c(2, 5)),
  array(c(letters, NA), c(3,3,3))
)
lst.ndim(x)

x <- list(
  1:10,
  array(1:10, 10),
  matrix(1:10, 2, 5),
  array(c(letters, NA), c(3,3,3))
)
dim(x) <- c(2,2)
dimnames(x) <- list(c("a", "b"), c("x", "y"))
lst.ndim(x)

```

rep_dim

*Replicate Array Dimensions***Description**

The `rep_dim()` function replicates array dimensions until the specified dimension sizes are reached, and returns the array.

The various broadcasting functions recycle array dimensions virtually, meaning little to no additional memory is needed.

The `rep_dim()` function, however, physically replicates the dimensions of an array (and thus actually occupies additional memory space).

Usage

```
rep_dim(x, tdim)
```

Arguments

`x` an atomic or recursive array or matrix.

`tdim` an integer vector, giving the target dimension to reach.

Value

Returns the replicated array.

Examples

```
x <- matrix(1:9, 3,3)
colnames(x) <- LETTERS[1:3]
rownames(x) <- letters[1:3]
names(x) <- month.abb[1:9]
print(x)

rep_dim(x, c(3,3,2)) # replicate to larger size
```

typecast

Atomic and List Type Casting With Names and Dimensions Preserved

Description

Type casting usually strips away attributes of objects.

The functions provided here preserve dimensions, dimnames, names, and [broadcaster](#) attributes, which may be more convenient for arrays and array-like objects.

The functions are as follows:

- `as_bool()`: converts object to atomic type logical (TRUE, FALSE, NA).
- `as_int()`: converts object to atomic type integer.
- `as_dbl()`: converts object to atomic type double (AKA numeric).
- `as_cplx()`: converts object to atomic type complex.
- `as_chr()`: converts object to atomic type character.
- `as_raw()`: converts object to atomic type raw.
- `as_list()`: converts object to recursive type list.

`as_num()` is an alias for `as_dbl()`.

`as_str()` is an alias for `as_chr()`.

See also [typeof](#).

Usage

```
as_bool(x, ...)
```

```
as_int(x, ...)
```

```
as_dbl(x, ...)
```

```
as_num(x, ...)
```

```
as_chr(x, ...)
```

```
as_str(x, ...)
```

```
as_cplx(x, ...)
```

```
as_raw(x, ...)
```

```
as_list(x, ...)
```

Arguments

`x` an R object.
`...` further arguments passed to or from other methods.

Value

The converted object.

Examples

```
# matrix example ====
x <- matrix(sample(-1:28), ncol = 5)
colnames(x) <- month.name[1:5]
rownames(x) <- month.abb[1:6]
names(x) <- c(letters[1:20], LETTERS[1:10])
print(x)
```

```
as_bool(x)
as_int(x)
as_dbl(x)
as_chr(x)
as_cplx(x)
as_raw(x)
```

```
#####
```

```
# factor example ====
x <- factor(month.abb, levels = month.abb)
names(x) <- month.name
print(x)
```

```
as_bool(as_int(x) > 6)
as_int(x)
as_dbl(x)
as_chr(x)
as_cplx(x)
as_raw(x)
```

Index

aaa00_broadcast_help, 2
aaa01_broadcast_operators, 4
aaa02_broadcast_casting, 7
acast, 7, 10
as_bool (typecast), 49
as_chr (typecast), 49
as_cplx (typecast), 49
as_dbl (typecast), 49
as_int (typecast), 49
as_list (typecast), 49
as_num (typecast), 49
as_raw (typecast), 49
as_str (typecast), 49
atomic, 25

bc.b, 5, 12
bc.b, ANY-method (bc.b), 12
bc.bit, 5, 13, 20
bc.bit, ANY-method (bc.bit), 13
bc.cplx, 5, 15
bc.cplx, ANY-method (bc.cplx), 15
bc.d, 5, 16
bc.d, ANY-method (bc.d), 16
bc.i, 5, 17
bc.i, ANY-method (bc.i), 17
bc.list, 6, 19
bc.list, ANY-method (bc.list), 19
bc.raw, 5, 20
bc.raw, ANY-method (bc.raw), 20
bc.rel, 5, 6, 12, 18, 20, 21
bc.rel, ANY-method (bc.rel), 21
bc.str, 5, 22
bc.str, ANY-method (bc.str), 22
bc_dim, 4, 24
bc_ifelse, 4, 25
bc_ifelse, ANY-method (bc_ifelse), 25
bc_strrep, 4, 26
bc_strrep, ANY-method (bc_strrep), 26
bcapply, 4, 23
bcapply, ANY-method (bcapply), 23
bind_array, 3, 27
broadcast (aaa00_broadcast_help), 2
broadcast-package
 (aaa00_broadcast_help), 2

broadcast_casting, 4, 11, 32, 33, 35, 40–43
broadcast_casting
 (aaa02_broadcast_casting), 7
broadcast_help (aaa00_broadcast_help), 2
broadcast_operators, 3, 13–15, 17–21, 23,
 24, 30, 31
broadcast_operators
 (aaa01_broadcast_operators), 4
broadcaster, 3, 5, 6, 25, 30, 49
broadcaster<- (broadcaster), 30

cast_dim2flat, 7, 32
cast_dim2hier, 7, 8, 33
cast_hier2dim, 7, 8, 34, 42, 43
cast_shallow2atomic, 7, 38
chol, 47
chol2inv, 47
cinv (linear_algebra_stats), 45

dropnests, 7, 8, 34, 40, 42

hier2dim, 7, 8, 34, 35, 42
hiernames2dimnames, 7, 8, 34, 35
hiernames2dimnames (hier2dim), 42

ifelse, 4, 25
in2out, 8

linear_algebra_stats, 45
list, 25
lst.ndim, 4
lst.ndim (ndim), 47

ndim, 4, 47

outer, 5

rep_dim, 4, 48

sd_lc (linear_algebra_stats), 45
simple linear algebra functions for
 statistics, 4
strrep, 4, 26

type-casting, 4

`typecast`, [49](#)

`typeof`, [49](#)

`undim(ndim)`, [47](#)